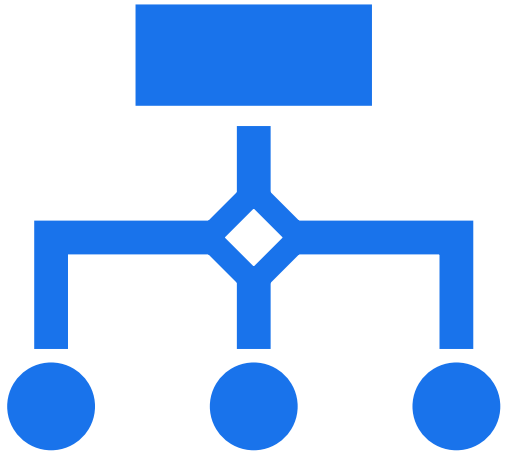


Actuators

YEAR 9 COMPUTING TECH

Learning Intention

- Describe inputs, storage, transmission, processes and outputs in mechatronic and automated systems



Success Criteria



I can identify and describe the purpose of actuators.



I can distinguish between robotic outputs

What is an Actuator?

Actuator: a component that helps machines to achieve physical movements by converting energy, often electrical, air, or hydraulic, into mechanical force. Simply put, it is the component in any machine that enables movement



Mechanical device that produces movement

Acts as an **output device** in mechatronic systems

Converts commands into physical actions

Connects the control system to the real world

Role of Actuators

Perform the **physical work** of a system

Receive signals from a microcontroller

Turn electrical signals into motion

Often described as the system's "muscles"

Types of Actuation



Electrical – powered by electricity (most common)



Hydraulic – powered by liquid pressure

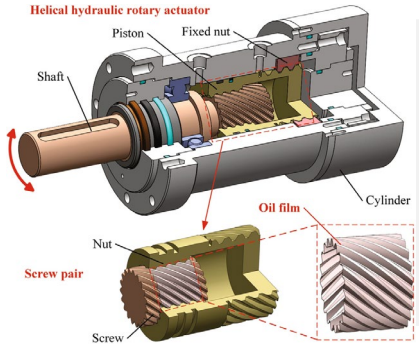


Pneumatic – powered by air pressure

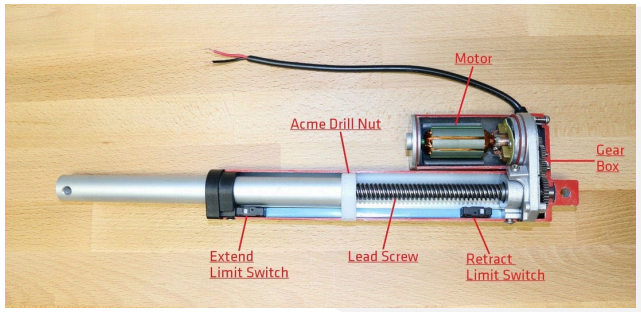


Used depending on required force and environment

Common Actuator Movements



Rotational motion (most common) – e.g. powering windshield wipers



Linear motion (back-and-forth movement) - think robotic arms

Motion can be converted using mechanisms:

Crank and slider

Rack and pinion

DC Motors



Provide **continuous rotation**

Simple and widely used

Found in toys, fans, and basic machines

Speed controlled by voltage

Servo Motors

Servo: a rotary or linear actuator that allows for precise control of angular or linear position, velocity, and acceleration in a mechanical system

Move to a **specific angle** (e.g. 0–180°)

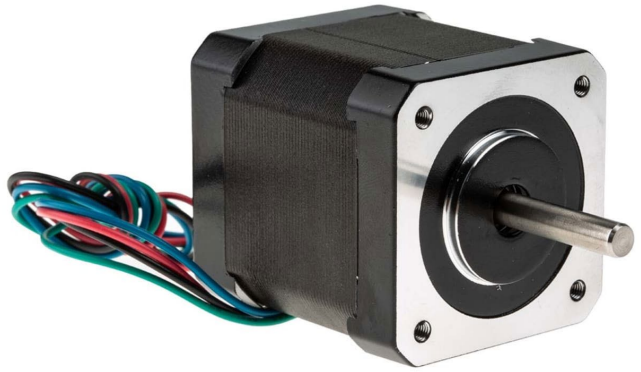
Use **feedback (closed-loop control)**

Hold position once reached

Used in robotics, steering systems, cameras



Stepper Motors



Move in **precise steps**

No feedback required

Highly accurate positioning

Used in **3D printers** and CNC machines

Other Actuators



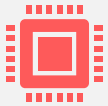
Solenoids – push/pull movement using magnetism



Relays – act as electrically controlled switches



Pumps – move liquids/gases to create motion



Used in valves, automation, and control systems

Real-World Applications



End Effector: devices attached to the end of a robot's arm to help it interact with the surrounding environment

- Automatic doors and windows
- Robotic arms, including end effectors
- Washing machines and valves
- Industrial automation and manufacturing